

Series 3: Applied Machine Learning and Predictive Modelling 1

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Load the `x-ray` dataset (file “Xray.RDS”). This study refers to a “study carried out to investigate the connection between X-ray usage and acute myeloid leukemia in childhood” (for further information see the help page about the `amlxray` dataset in `{faraway}` package). We work on a subset of the original data, in particular, we consider the following variables:

- `disease`: is the response variable (1 = *diseased*, 0 = *healthy*)
- `Sex`: Sex of the patient
- `age`: Age of the child (in years)
- `downs`: Presence of the Down syndrome
- `Mray`: Did the mother ever had an Xray
- `Fray`: Did the father ever had an Xray
- `CnRay.num`: Total number of Xrays of the child "1" = none; "2" = 1 or 2; "3" = 3 or 4; "4" = 5 or more

```
d.xray <- readRDS("../..//Datasets/Xray.RDS")
str(d.xray)
```

```
tibble [112 x 7] (S3: tbl_df/tbl/data.frame)
 $ Sex      : Factor w/ 2 levels "F","M": 1 2 1 2 2 1 1 2 1 2 ...
 $ disease  : num [1:112] 1 1 0 1 1 1 1 0 0 0 ...
 $ age      : int [1:112] 0 6 8 1 4 9 17 5 0 7 ...
 $ downs    : Factor w/ 2 levels "no","yes": 1 1 1 2 1 1 1 1 1 1 ...
 $ Mray     : Factor w/ 2 levels "no","yes": 1 1 1 1 2 2 1 1 1 1 ...
 $ Fray     : Factor w/ 2 levels "no","yes": 1 1 1 1 1 1 1 1 1 1 ...
 $ CnRay.num: num [1:112] 1 3 1 1 1 1 2 1 1 1 ...
```

```
head(d.xray)
```

```
# A tibble: 6 x 7
  Sex  disease  age downs Mray  Fray  CnRay.num
<fct> <dbl> <int> <fct> <fct> <fct>    <dbl>
1 F      1      0 no    no    no        1
```

2	M	1	6	no	no	no	3
3	F	0	8	no	no	no	1
4	M	1	1	yes	no	no	1
5	M	1	4	no	yes	no	1
6	F	1	9	no	yes	no	1

Question

Visualise the data. You may want to take advantage of the `pairs()` function.

Question

Fit a starting model to the data. Start by assuming linearity for the continuous predictors and assume that no interaction is needed. Comment on the results obtained with the `summary()` function.

Question

Interpret the coefficients of `SexM` and `age`

Question

Look at the estimated effect of the variable `downs` and its p-value. Do the results make sense to you? Comment.

Question

Ask generative AI to

- *explain the differences between binomial and binary data, and provide some examples*
- *explain what model you should fit to binary data*
- *explain what happens when you fit a normal linear model to binary data*
- *interpret the coefficients you get from a logistic regression*

What can you share about these answers?